

MATH 263 Notes

Chapter 2 - Logic and Statements

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2.1 - Logical Form and Logical Equivalence

In this class, we are going to deal with different types of statements and arguments. To simplify things and make them more generalizable, we often use symbols.

A common form of argument is

If p , then q .

This is called a *conditional sentence*, where p is the hypothesis and q is the conclusion. An example of a conditional sentence is

If it is snowing, then it is cold.

where p is “it is snowing” and q is “it is cold”.

For now, we will use these three symbols:

- **and:** $\wedge$

For $p \wedge q$ to be true, both p and q *must* be true.

- **or:** $\vee$

For $p \vee q$ to be true, at least one of p or q must be true.

- **not:** $\neg, \sim$

If p is true, $\neg p$ is false.

*In some coding languages like Java and C++, negation is denoted with !

If you have a sentence with a combination of these, order of operations says negation comes first.

A statement is a sentence that is either true or false, but it cannot be both. We say that two statements are *logically equivalent* if and only if they have identical truth values for every possible substitution.

One way to view all the truth values is through a truth table. We put one column for each

variable, and then one column for each expression we want to evaluate. The goal is to see the truth value of a statement for every possible combination of truth values for the variables. If two statements have identical columns, then we can say they are logically equivalent.

Example 1 Truth tables for $p \wedge q$ and $p \vee q$:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

De Morgan's Laws:

- $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Example 2 Proving DeMorgan's Law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

p	q	$\neg(p \wedge q)$	$\neg p \vee \neg q$
T	T	F	F
T	F	T	T
F	T	T	T
F	F	T	T

A *tautology* is a statement that is always true, no matter what variables are plugged in. An example of this is $p \vee \neg p$ (you can verify this with a truth table!). A *contradiction* is a statement that is always false, no matter what variables are plugged in. An example of this is $p \wedge \neg p$.

2.2 - Conditional Statements

Sentences of the form “If p , then q ” are called *conditional*, where p is the hypothesis, or premise, and q is the conclusion. We can also say that “ p implies q ” and denote this with $p \implies q$.

The sentence $p \implies q$ is only false when the hypothesis is true and the conclusion is false. A false premise can lead to any sort of conclusion, which is why both $F \implies T$ and $F \implies F$ are true.

Below is the truth table.

p	q	$p \implies q$
T	T	T
T	F	F
F	T	T
F	F	T

A conditional statement where the premise is false is called *vacuously true*. These statements are true, no matter what the conclusion is.

The order of operations for statements that include $\implies$, $\wedge$, $\vee$, and $\neg$ is (from first to last):

1. $\neg$
2. $\wedge$, $\vee$ (from left to right)
3. $\implies$

The negation of a conditional statement is

$$\neg(p \implies q) \equiv p \wedge \neg q.$$

So, the negation of “if p , then q ” is “ p and not q ”.

Example 3

Let s be the conditional sentence “if it snows, then I will not go to school.” The negation of s then is “it is snowing and I will go to school.”

*Notice that the negation of a conditional statement does **not** include “if”.

The *contrapositive* of a conditional statement $p \implies q$ is

$$\neg q \implies \neg p.$$

In words, the contrapositive of “If p , then q ” is “If not q , then not p ”.

A conditional statement and its contrapositive are logically equivalent.

Given a conditional statement of the form “If p , then q ” ($p \implies q$):

- The *converse* is “If q , then p ” ($q \implies p$)
- The *inverse* is “If not p , then not q ” ($\neg p \implies \neg q$)

These are **not** logically equivalent to “If p , then q ”, but the converse and inverse are logically equivalent to each other.

Biconditional

The *biconditional* of p and q is “ p if and only if q ”, and is denoted by $p \iff q$. You can think of it as $(p \implies q) \wedge (q \implies p)$. Below is the truth table for $p \iff q$.

p	q	$p \iff q$
T	T	T
T	F	F
F	T	F
F	F	T

Necessary and Sufficient Conditions

Let p and q be statements.

- p is a *sufficient condition* for q means “if p , then q ”.
- p is a *necessary condition* for q means “if not p , then not q ”
 - The contrapositive of “if not p , then not q ” is “if q , then p ”. Since a statement and its contrapositive are equivalent, we have that
“ p is a *necessary condition* for q ” also means “if q , then p ”

Consequently, p is a necessary and sufficient condition for q means “ p if and only if q ”.

2.3 - Valid and Invalid Arguments

An *argument* is a sequence of statements. In an argument, the last statement is the conclusion and the remaining statements are premises.

An *argument form* is the abstract form of an argument.

For example, if the argument is

If it is snowing, then it is cold.

It is snowing.

Therefore, it is cold.

Then the abstract form is

If p , then q .

p

$\therefore q$.

We say that an argument form is *valid* if, when all of the premises are true, the conclusion is true. To determine if an argument is valid, we look at its form - if its form is valid, then the argument is valid.

An argument form consisting of two premises and a conclusion is called a *syllogism*.

A *rule of inference* is a form of argument that is valid. Below are some of the rules of inference (these are not all of them).

1. Modus ponens

$p \implies q$

p

$\therefore q$.

2. Modus tollens

$p \implies q$

$\neg q$

$\therefore \neg p$.

3. Generalizations

p

$\therefore p \vee q$.

4. Specialization

$p \wedge q$

$\therefore p$.

5. Elimination

$$p \vee q$$

$$\neg q$$

$$\therefore p.$$

6. Transitivity

$$p \implies q$$

$$q \implies r$$

$$\therefore p \implies r.$$

7. Division into cases

$$p \vee q$$

$$p \implies r$$

$$q \implies r$$

$$\therefore r.$$

8. Contradiction - Let c be a contradiction.

$$\neg p \implies c$$

$$\therefore p.$$

If an argument has a form where all of the premises are true and the conclusion is false, then it is an *invalid argument*.

A *fallacy* is an error in reasoning that results in an invalid argument.

1. Converse error - This starts with a conditional statement, then says that the conclusion is true, and concludes that the premise must then be true.

$$p \implies q$$

$$q$$

$$\therefore p$$

2. Inverse error - This fallacy incorrectly assumes that the inverse of a conditional statement is logically equivalent to the original statement.

$$p \implies q$$

$$\neg p$$

$$\therefore \neg q$$

An argument is called *sound* if, and only if, it is valid and all its premises are true. An argument that is not sound is called *unsound*.